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26 July 1979
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15 January 1975

MILITARY SPECIFICATION

CONVERTER-PATCH PANEL CV-3022/UG

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance, design, test, manufacture, and acceptance requirements for the converter-patch panel CV-3022/UG. The converter-patch panel is a signal level converting device between interface (low-level) and interim standard interface (high-level, neutral, 60-milliampere (mA)) signals as described by MIL-STD-188.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue listed in the Department of Defense Index of Specifications and Standards (DoDISS) and its supplements, form a part of this document to the extent specified herein. The date of the applicable DoDISS and supplements thereto shall be as specified in the solicitation.

SPECIFICATIONS

MILITARY

MIL-S-901	Shock Tests, H.I. (High-Impact), Shipboard Machinery, Equipment And Systems, Requirements For
MIL-E-16400	Electronic, Interior Communication And Navigation Equipment, Naval Ship And Shore, General Specification For
MIL-E-17555	Electronic And Electrical Equipment, Accessories, And Repair Parts; Packaging And Packing Of

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-109	Quality Assurance Terms And Definitions
MIL-STD-167-1	Mechanical Vibrations Of Shipboard Equipment (Type I - Environmental And Type II - Internally Excited)
MIL-STD-188	Military Communication System Technical Standards
MIL-STD-188-100	Common Long Haul And Tactical Communication System Technical Standards
MIL-STD-454	Standard General Requirements For Electronic Equipment

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command (ELEX-5043), Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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MIL-STD-461	Electromagnetic Interference Characteristics Requirements For Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement Of
MIL-STD-471	Maintainability Demonstration
MIL-STD-781	Reliability Design Qualification And Production Acceptance Tests: Exponential Distribution
MIL-STD-810	Environmental Test Methods
MIL-STD-965	Parts Control Program
MIL-STD-1399, Section 300	Interface Standard For Shipboard Systems, Section 300, Electric Power, Alternating Current (Metric)

HANDBOOKS

MILITARY

MIL-HDBK-241	Design Guide For EMI Reduction In Power Supplies

DRAWINGS

Naval Electronic Systems Command

28687 - DL 71E2N200	Data List, Converter-Patch Panel, CV-3022/UG
28687 - 71E2N200	Converter-Patch Panel CV-3022/UG 1A1
28687 - 71E2N201	Installation Drawing, Converter-Patch Panel CV-3022/UG
28687 - 71E2N202	Electrical Schematic, Converter-Patch Panel CV-3022/UG
28687 - 71E2N204	Patchcord, 2-Conductor Shielded
28687 - 71E2N296	Relay, Telegraph, Passive Solid-State (Low-to-High Level)
28687 - 71E2N297	Relay, Telegraph, Passive Solid-State (High-to-Low Level)
28687 - 71E2N460	Front Panel Assembly Converter(1A2)

(Copies of specifications, standards, handbooks, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. The converter-patch panel shall conform to the requirements of MIL-E-16400, except as hereinafter specified.

3.1.1 First article. When specified (see 6.2), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.3).

3.1.2 Converter-patch panel description. The converter-patch panel CV-3022/UG shall provide a signal level interface of seven channels between the interim standard interface (high-level) and the standard interface (low-level) of direct current (DC) digital transmission reference systems of military communications equipment described in MIL-STD-188-100. The converter-patch panel also provides 14 channels for low-level signal patching capability.

3.2 Characteristics.

3.2.1 Performance characteristics. The converter-patch panel shall consist of the circuits specified in 3.2.1.1 and 3.2.1.2 and a self-contained power supply supplying an unregulated ±14 volts (V) and a regulated ±6.5 V. The circuits shall be as shown on drawing (DWG) 28687 - 71E2N202.

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3.2.1.1 High-to-low level signal converter circuit. The high-to-low level signal converter circuits shall be capable of converting an interim standard interface signal (high-level, 130 V, neutral, 60 mA) to the standard interface signal (low-level) for digital DC communication equipment described in MIL-STD-188, except that pulse rise and fall times are fixed at 250 microseconds (μs) ±20 percent. There shall be seven identical circuits for channel 0 through channel 6 using solid-state relays (DWG 28687 - 71E2N297). Input-to-output signal isolation shall be a minimum of 50 megohms. The high-to-low level circuit input shall be capable of accepting signals which conform to an interim standard interface (high-level, 130 V, neutral, 60 mA) of MIL-STD-188-100. Each channel input shall have the capability to provide either signal loop current to, or sink loop current from, peripheral equipment. The equipment shall also contain a means by which loop current can be monitored and varied between 40 to 80 mA directly from the front panel. The high-to-low level circuit output shall be capable of producing outputs for the standard interface low-level requirements of MIL-STD-188-100, except as noted herein.

3.2.1.2 Low-to-high level signal converter circuit. The low-to-high level signal converter circuits shall be capable of converting the standard interface (low-level) to the interim standard interface (high-level) (for example, the reverse of 3.2.1.1). There shall be seven identical circuits for channel 0 through channel 6 using solid-state relays (DWG 28687 - 71E2N296). The low-to-high level signal converter circuit shall be capable of accepting signals which conform to the standard interface (low-level) requirements of MIL-STD-188-100. The low-to-high level signal converter circuit shall produce outputs for the interim standard interface (high-level, 130 V, neutral, 60 mA) of MIL-STD-188-100. Each channel output shall have the capability to provide either signal loop current to, or sink loop current from, peripheral equipment. The equipment shall also contain a means by which loop current can be monitored and varied between 40 to 80 mA directly from the front panel.

3.2.1.3 Patching capability. The patching sections shall be a part of the front panel of the equipment and shall be accessible from the front panel. The sections shall be capable of patching 14 channels using standard two-section phone jacks and plugs. The patching circuits shall be capable of patching only standard interface (low-level signals described in MIL-STD-188-100).

3.2.1.4 Patching circuits. There shall be four phone jacks per channel in each of the patching circuits, one monitor jack, one line jack, one equipment jack, and one miscellaneous jack. The line, equipment, and miscellaneous jacks per channel shall be directly connected to the input/output (I/O) connectors, A1J7, A1J8, A1J9, A1J10, A1J11, and A1J12 on the rear panel. The monitor jacks shall have the same signal as the line jacks without any rear panel connection. The line jack signal shall be connected to the equipment jack and to the connectors without patching. Placing a phone plug in the line jack, a 6.0 ±1.0 VDC signal shall be present on the equipment jack output. The miscellaneous jacks shall be independent of the other jacks and will be utilized for other functions as desired.

3.2.2 Physical characteristics.

3.2.2.1 Weight limit. The total weight of the converter-patch panel, excluding cables, shall not exceed 22.72 kilograms (kg) (50 pounds (lbs)).

3.2.2.2 Interface requirements. The equipment shall conform to the installation control and interface requirements of DWG 28687 - 71E2N201.

3.2.2.3 Standard conditions. The following conditions shall be used as standard conditions to establish normal operating performance characteristics:

a. Temperature. Room ambient (25° Celsius (C), ±2°C
b. Atmospheric pressure. 28.5, +2.0, -4.5, inches of mercury
c. Relative humidity (RH). 50 percent ±30 percent
d. Primary input power. 60 hertz (Hz) single-phase 115 VAC
e. Secondary input power. DC voltage 130 VDC

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3.2.3 Reliability requirements.

3.2.3.1 Quantitative reliability requirement. The specified upper test mean-time-between-failures (MTBF) (θ_0 as defined by MIL-STD-781) shall be 1000 hours for the converter patch panel. The specified lower test MTBF (θ_1 as defined by MIL-STD-781) shall be 500 hours.

3.2.4 Environmental conditions. The converter-patch panel shall conform to the performance requirements specified herein with no degradation when subjected to the environmental service conditions of MIL-E-16400 for Class 4 equipment. When the item is operating under these conditions, none of the following defects shall occur as a result of these conditions:

a. No loosening of fixed parts or subassemblies
b. No moving part shall become loose or sluggish when operating.
c. No moveable control shall shift from its setting except as noted. It is allowable for circuit breaker A2CB1, part of front panel assembly, DWG 28687 - 71E2N460, to trip under shock conditions.

3.2.4.1 Shock and vibration. The converter-patch panel shall withstand the shock and vibration test requirements specified in MIL-E-16400 for shipboard and amphibious electronic equipment (Class 4).

3.2.4.2 Service shock environment. The equipment shall operate as specified herein after being subjected to the bench handling testing of 4.4.3.2.

3.2.4.3 Temperature and humidity. The temperature and humidity shall be in accordance with MIL-E-16400, Class 4.

3.2.5 Warm-up time. The warm-up time required for the converter-patch panel, prior to its operation, shall not exceed one minute under standard and extreme service conditions.

3.2.6 Operating power. The operating power requirements shall be single-phase alternating current (AC) power, 115 V, 115 volt-amperes (VA), 60 Hz. The converter-patch panel shall conform to the specified performance characteristics when connected to a Type 1 shipboard power system, as specified in MIL-STD-1399, Section 300. The equipment also requires 130 VDC ±10 percent power when used to provide signal loop current for interim standard interface (high level, 130 VDC, neutral, 60 mA) channels at 60 mA per channel.

3.3 Design and construction.

3.3.1 Production drawings. The converter-patch panel shall be fabricated and assembled in accordance with the drawings, parts lists, and other documents listed on Data List 28687 - DL 71E2N200.

3.3.2 Selection of parts and materials. The parts and materials used in the converter-patch panel shall be as specified in 3.3.1.

3.3.2.1 Parts control. The parts to be incorporated in the equipment shall be controlled in accordance with MIL-STD-965, Procedure I.

3.3.3 Electromagnetic compatibility. The converter-patch panel shall be self-compatible and shall conform to the requirements of MIL-STD-461 for Class IC equipment, except as follows:

	Method	Frequency	Emission type	Relaxation (dB)
a.	CE03	17 MHz	NB	2
b.	CE03	16 MHz	BB transients	23 1/
c.	CE01	950 Hz	NB	10

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	Method	Frequency	Emission type	Relaxation (dB)
d.	REO2	6 MHz	BB	15
e.	REO2	15 MHz	BB transients	27 1/
f.	RSO1	4 to 25 kHz	NB	-50 2/
g.	All others, no change			

1/ On-off switching transients

2/ The magnetic field susceptibility requirement is increased due to the field of co-located equipment

3.3.4 Interchangeability. Interchangeability shall be in accordance with Requirement 7 of MIL-STD-454.

3.3.5 Safety. The patch-panel converter shall conform to Requirement 1 of MIL-STD-454.

3.3.6 Leakage current. The leakage current from the enclosure shall not exceed 5 mA to ground.

3.3.7 Workmanship.

3.3.7.1 General workmanship. Workmanship shall conform to Requirement 9 of MIL-STD-454 and DL 71E2N200 (see 3.3.1).

3.3.7.2 Workmanship screen. All equipment produced in accordance with this specification shall withstand a defect detection vibration screen of random type vibration at $0.04g^2/Hz$ ±3dB from 80 Hz to 350 Hz and temperature cycling as specified in 4.4.1.2.3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of, but is not limited to, a) Surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied, b) Government product inspection to measure quality of the product to be offered for acceptance, c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

4.1.2 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.1.3 Standard conditions. Unless otherwise specified herein, testing shall be conducted under the conditions specified in 3.2.2.3.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)
   1. In-process inspection (see 4.4.1)
   2. Production inspection (see 4.4.2)
   3. Environmental inspection (see 4.4.3)
c. Reliability qualification testing (see 4.5)
d. Maintainability demonstration (see 4.6)
e. Inspection of preparation for delivery (see 4.7)

4.3 First article inspection. Unless otherwise specified (see 6.2), one unit shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the examination and testing specified in 4.4.2. and 4.4.3 and the additional first article inspections listed in TABLE I.

TABLE I. Additional first article inspections.

Inspection or test	Requirement paragraph	Test paragraph
Shock	3.2.4.1	4.3.1
Vibration	3.2.4.1	4.3.2
Patching circuits	3.2.1.4	4.4.2.2
High-to-low level signal converter circuit	3.2.1.1	4.4.2.2
Low-to-high level signal converter circuit	3.2.1.2	4.4.2.2
Leakage current	3.3.6	4.4.2.3
Operating power	3.2.6	4.4.2.1
Temperature (Level 4)	3.2.4.3	MIL-E-16400
Humidity	3.2.4.3	MIL-E-16400
Service shock	3.2.4.2	4.4.3.2
Electromagnetic interference	3.3.3	4.4.3.3

4.3.1 Shock. Shock testing shall be conducted in accordance with MIL-S-901, Type A, for Class 1, lightweight equipment, and shall conform to the requirements of 3.2.4.1.

4.3.2 Vibration. Vibration testing shall be conducted in accordance with MIL-STD-167-1 for Type I equipments and shall conform to the requirements of 3.2.4.1.

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4.4 Quality conformance inspection.

4.4.1 In-process inspections. Inspections shall be performed in accordance with the quality assurance provisions required in the appropriate item drawing as listed in DL 71E2N200. The in-process inspections shall be satisfactorily completed prior to beginning production inspection.

4.4.1.1 General workmanship inspection. The equipment, including subassemblies and assemblies, shall be examined for workmanship and soldering during the fabrication and assembly process for conformance to the requirements of 3.3.7.1.

4.4.1.2 Vibration test. Prior to conducting the temperature cycling test, a vibration test shall be performed on each equipment to verify compliance with the requirements of 3.2.4.1. The vibration may be performed at the module, drawer, or end item level. All the hardware, including cables and connectors, shall be exposed to vibration.

4.4.1.2.1 The vibration shall be random, or subject to procuring activity approval, pseudo-random or complex waveform vibration, for an accumulated time of 10 minutes in the axis deemed most susceptible to vibration excitation. All test items shall be hard-mounted (without shock isolators) and subjected to the vibration conditions of FIGURE 1. The control accelerometer shall be located next to one of the mounting points of the item under test. Equipment having a bandwidth no greater than 10 Hz for vibration frequencies up to 500 Hz and 100 Hz for vibration frequencies above 500 Hz shall be used for the control and analysis of the acceleration spectral density.

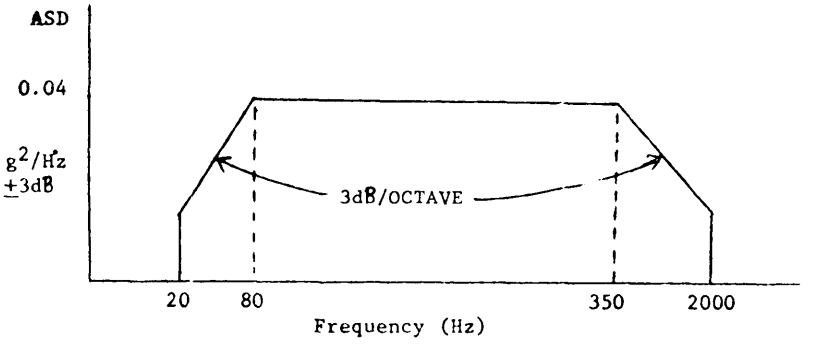


FIGURE 1. Random vibration curve.

4.4.1.2.2 The test item shall be energized during vibration and appropriate input signals applied to observe any abnormal conditions of the output functional characteristics. All failures occurring during test shall be corrected and the test resumed.

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4.4.1.2.3 Temperature cycling. Each equipment shall be subjected to the temperature curve shown in FIGURE 2. The number of cycles required shall be determined by the electrical and electronic parts count and the parts count chart shown herein. The temperature rate of change shall not be less than 5°C per minute. Equipment power shall be turned on and off at the indicated times. When practical, the equipment drawers, panels, and enclosures shall be opened or removed for maximum exposure to the changing temperature. The last temperature cycle shall be failure-free. When performance measurements are called for, a functional operating test shall be performed. The dwell time shall be 80 percent of the time required for the largest electrical or electronic part to become temperature stabilized. Temperature stabilization shall be as defined in MIL-STD-810 for the non-operating condition.

Electric and electronic parts count	No. of temperature cycles
500 or less	3
501 to 1000	4
1001 to 2000	6
2001 to 3000	8
3001 and up	10

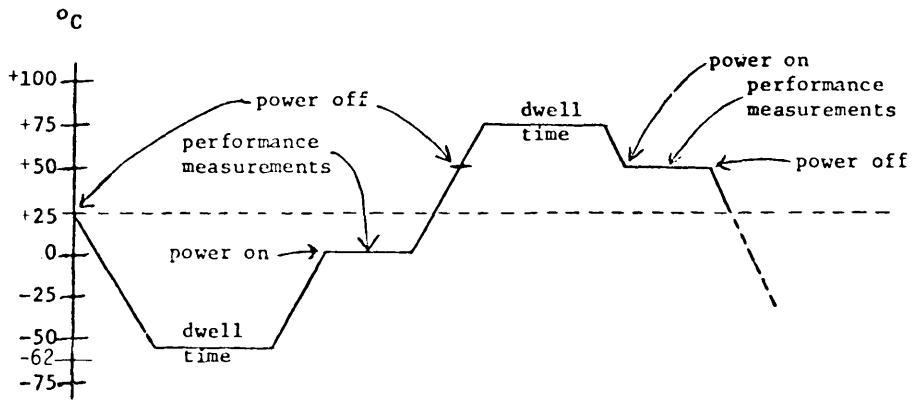


FIGURE 2. Temperature curve (one cycle).

4.4.1.2.4 Failures during temperature cycling. Failures occurring prior to the last original temperature cycle shall be repaired and the cycling continued. If failures occur during the last original temperature cycle, the chart shown in TABLE II shall be used to determine the number of additional cycles required.

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TABLE II. Failures during temperature cycling.

Percentage of total parts reworked and replaced 1/	Equipment parts count		
	500 or less	501 to 2000	2001 and up
Up to 0.1 percent	None	Repeat 1 cycle	Repeat 1 cycle
0.1 percent to 1 percent	Repeat 1 cycle	Repeat 1 cycle	Repeat 2 cycles
1 percent to 5 percent	Repeat 1 cycle	Repeat 2 cycles	Repeat 4 cycles
5 percent to 10 percent	Repeat 2 cycles	Repeat 4 cycles	Repeat 6 cycles
	Last cycle shall be failure-free		

1/ Each reworked solder joint or resoldered lead shall count as one part replaced.

4.4.2 Production inspection. Production inspection, including sampling, shall conform to TABLE III and to the inspection procedures of MIL-STD-105 using general inspection Level II. The equipment shall satisfactorily conform to the requirements of production inspection prior to release for shipment. Rejected lots shall be reinspected using tightened inspection only after each equipment in the lot has been reinspected for the failed attributes.

TABLE III. Production inspection.

Inspection or test	Requirement paragraph	Test paragraph	AQL percent
Patching circuits	3.2.1.4	4.4.2.2	1.5
High-to-low level signal converter circuits	3.2.1.1	4.4.2.2	1.5
Low-to-high level signal converter circuit	3.2.1.2	4.4.2.2	1.5
Leakage current	3.3.6	4.4.2.3	1.5
Operating power	3.2.6	4.4.2.1	6.5

4.4.2.1 Operating power.

4.4.2.1.1 Steady-state voltage and frequency. The steady-state voltage and frequency test shall be performed in accordance with the steady-state voltage and frequency paragraph of *MIL-E-16400*.

4.4.2.1.2 Transient voltage. Transient voltages shall be as follows:

a. Upper limit. With the equipment operating in the upper limit of steady-state voltage, a transient voltage of +18 percent of normal voltage recovering to the steady-state voltage within two seconds shall be superimposed. The equipment shall be capable of normal operation following the transient.

b. Lower limit. With the equipment operating on the lower limit of steady-state voltage, transient voltage of -18 percent of normal voltage recovering to the steady-state voltage within two seconds shall be superimposed. The equipment shall be capable of normal operation following the transient.

c. Transient (700 V). In addition to these tests, equipments containing transistors or semiconductor devices shall be subjected to an input supply line transient of 700 V amplitude and two-millisecond (ms) duration while the equipment is operating at normal supply line voltage and frequency. The equipment shall be capable of normal operation immediately following the transient.

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4.4.2.1.3 Transient frequency. Transient frequencies shall be as follows:

a. Upper limit. With the equipment operating at five percent above normal frequency, the frequency shall be increased by an additional three percent recovering to the steady-state frequency (±5 percent of normal) within two seconds. The equipment shall be capable of normal operation following the transient.
b. Lower limit. With the equipment operating at five percent below the normal frequency, the frequency shall be decreased by an additional three percent, recovering to the steady-state frequency (±5 percent of normal) within two seconds. The equipment shall be capable of normal operation following the transient.

4.4.2.1.4 Spike voltage test. In addition to the test specified in 4.4.2.1.3, the equipment shall be subjected to an input supply line voltage spike of 2500 V positive peak amplitude; the wave shape shall correspond with that of the figure for the spike voltage (short time transient) wave shape, Section 300 of MIL-STD-1399. This spike shall be impressed at normal supply line voltage and frequency while the equipment is in operation. The equipment shall be capable of normal operation immediately following the test.

4.4.2.1.5 Power interruption. With the equipment operating within the steady-state tolerances of voltage and frequency, the 115 VAC power shall be suddenly interrupted, and after an interval between three and four seconds, the power supply, within the steady-state tolerances, shall be reapplied. After the equipment has been operated long enough to detect any major performance degradation and to include any recycling time, the power shall be interrupted for an interval of 29 to 30 seconds. The cycle, consisting of three such interruptions, shall be repeated for a total of four times. During these tests, and as a result of these tests, no damage to the equipment shall be incurred, and observed effects of power interruption or reapplication shall fall within the performance tolerances of the individual equipment specification.

4.4.2.1.6 Power and power factor. Measurements of input KVA, power, and power factor shall be made at normal line voltage and frequency on each primary supply line to each unit of the equipment, including each phase of three-phase supplies, under each principal mode of operation of the equipment to determine compliance with power requirements specified herein. During any mode of equipment operation which imposes cyclic and random load changes on the primary supply line, the input power variation shall be measured to determine conformance with the requirements of 3.2.6.

4.4.2.2 Operating tests. The equipment, assembled to include all parts required for delivery of end items, shall be subjected to the following operating test:

a. High-to-low converter test, source mode. Place the hi-to-lo converter loop current select switch (A3S1) in the source position. Connect a switch to the hi-in connector A1J3 pins A (loop out) and B (loop in) capable of conducting a 100 mA current and possessing an effective DC series resistance of 50 ±2 ohms. Connect a 6000 ohm ±5 percent, 0.25 watt load resistor across lo-out connector A1J5 pin A (signal) and B (return). With the switch closed and conducting a current of 60 ±4 mA, the output signal on A1J5 shall be +6.0 ±1.0 VDC. With the switch open, the output signal on A1J5 shall be -6.0 ±1.0 VDC. With the switch opening and closing at a rate of 37 ±2 Hz, the output signal shall conform to MIL-STD-188, except the signal rise and fall time requirements shall be 250 μs ±20 percent when measured between the 10 percent and 90 percent amplitude points.
b. High-to-low converter test, sink mode. With the hi-to-lo converter loop current select switch (A3S1) in the sink position and a 130 VDC power supply placed in series with the switch of a as shown in FIGURE 3, the signal output requirements of a shall be present on the lo-out connector pins A (signal) and B (return).
c. Repeat the procedure of a and b for channels 1 through 6 with the applicable connections shown in TABLE IV.

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d. Low-to-high converter test, source mode. Place the lo-to-hi converter loop current select switch (A3S8) in the SOURCE position. Connect a 50 ±2 ohm, five watt load resistor between hi-out connector A1J6 pin A (loop out) and B (loop in). Apply a standard interface (low level, ±6 volts) signal to lo-in connector A1J4 pin A (signal) and pin B (return). The standard interface input signal shall have a repetition rate in switching from +6 V to -6 V to +6 V of 37 ±2 Hz. When a 6.0 ±1 VDC signal is applied to A1J4 pin A, a 60 ±4 mA current shall eminate from connector A1J6 pin A, pass through the 50 ±2 ohm load resistor and return to the unit through connector A1J6 pin B. When a -6.0 ±1 VDC signal is applied to A1J4 pin A, the output 50-ohm load resistor shall not conduct any current.

e. Low-to-high converter test, sink mode. With the lo-to-hi converter loop current select switch (A3S8) in the SINK position, hi-out connector A1J6 pins A and B terminated with the load shown in FIGURE 4 and applying a standard interface (low level) signal to connector A1J4 pins A (signal) and B (return), the output signal requirements specified in d shall be present on A1J6.

f. Low-to-high converter retest. Repeat the procedures of d and e for channels 1 through 6 with the applicable connections shown in TABLE V.

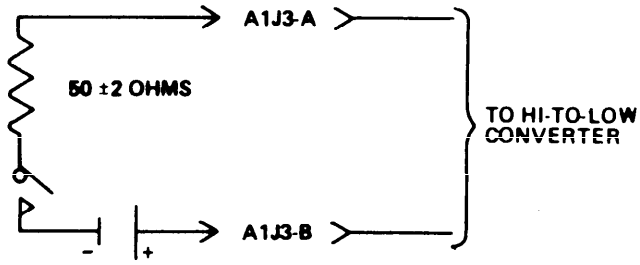


FIGURE 3. Input Circuit, sink mode.

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TABLE IV. High-to-low converter channel controls.

Channel	Loop current		Hi-in connector A1J3		Lo-out connector A1J5	
	Select switch	Adjust control	Loop-in	Loop-out	Signal	Return
0	A3S1	A2R1	B	A	A	B
1	A3S2	A2R2	E	D	D	E
2	A3S3	A2R3	H	G	G	H
3	A3S4	A2R4	L	K	K	L
4	A3S5	A2R5	P	N	N	P
5	A3S6	A2R6	T	S	S	T
6	A3S7	A2R7	V	U	U	V

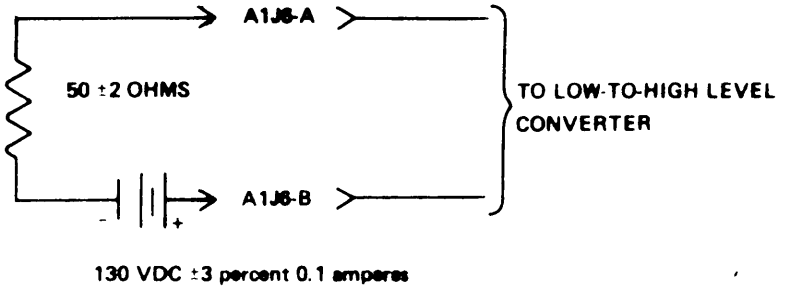


FIGURE 4. Output load, sink mode.

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TABLE V. Low-to-high converter channel controls.

Channel	Loop current		Lo-in connector A1J4		Hi-out connector A1J6	
	Select switch	Adjust control	Signal	Return	Loop-in	Loop-out
0	A3S8	A2R8	A	B	B	A
1	A3S9	A2R9	D	E	E	D
2	A3S10	A2R10	G	H	H	G
3	A3S11	A2R11	K	L	L	K
4	A3S12	A2R12	N	P	P	N
5	A3S13	A2R13	S	T	T	S
6	A3S14	A2R14	U	V	V	U

g. Patching circuit line tests. Apply a standard interface (low level) signal as defined by MIL-STD-188-100 to line connector A1J7 pins A (signal) and B (ground). The input signal on connector A1J7 shall have a repetition rate between 75 and 2400 baud. The input signal,as applied to A1J7, shall be present on EQUIP connector A1J9 pins A (signal) and B (ground).

h. Monitor patching. Using the patching cable, W1, DWG 28687 - 71E2N204, patch the signal from SECTION A MON CHANNEL 0 jack, A2J1 to SECTION A MISC CHANNEL 0 jack, A2J43. The input signal applied to line connector, A1J7, shall be present on MISC connector A1J11 pins A (signal) and B (ground).

i. LINE-to-MISC patching. Remove the patching cable from the SECTION A CHANNEL 0 MON jack, A2J1 and place the patching cable in the SECTION A CHANNEL 0 LINE jack A2J15. The input signal applied to line connector A1J11 pins A (signal) and B (ground). An output signal of +6.0 $\pm$1 VDC shall be present on EQUIP connector A1J9 pins A (signal) and B (ground).

j. Patching retest. Repeat the procedure of g, h, and i for channels 1 through 6 in sections A and B with the applicable connections as shown in TABLES VI and VII.

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TABLE VI. Section A.

Channel	Line connector A1J7 pins		Equip connector A1J9 pins		Misc connector A1J11 pins	
	Signal	Return	Signal	Return	Signal	Return
1	D	E	D	E	D	E
2	G	H	G	H	G	H
3	K	L	K	L	K	L
4	N	P	N	P	N	P
5	S	T	S	T	S	T
6	U	V	U	V	U	V

TABLE VII. Section B.

Channel	Line connector A1J8 pins		Equip connector A1J10 pins		Misc connector A1J12 pins	
	Signal	Return	Signal	Return	Signal	Return
1	D	E	D	E	D	E
2	G	H	G	H	G	H
3	K	L	K	L	K	L
4	N	P	N	P	N	P
5	S	T	S	T	S	T
6	U	V	U	V	U	V

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4.4.2.3 Leakage current test.

WARNING

THIS TEST MAY BE HAZARDOUS DUE TO THE UNGROUNDED CONDITION OF THE EQUIPMENT DURING THE TEST. DO NOT TOUCH EXPOSED METAL SURFACES.

THE UNITED STATES GOVERNMENT NEITHER ASSUMES NOR ACCEPTS RESPONSIBILITY FOR ANY INJURY OR DAMAGE THAT MAY OCCUR DURING OR AS A RESULT OF THIS TEST.

4.4.2.3.1 Equipment connections. Each equipment directly connected to an external power source and units deriving power from the equipment shall be placed on an insulated surface. All safety ground conductors between the equipment and units deriving power from the equipment shall be intact. The safety ground conductor between the equipment under test and the source power shall be opened during the test. The equipment shall be connected as shown in FIGURE 5 if it is connected to single-phase or DC source power. When DC measurements are performed do not reverse polarity unless the equipment has reverse polarity protection.

4.4.2.3.2 Measurement. Leakage current shall be measured on equipment in its normal operating configuration. Equipment controls in each operating mode shall be such that maximum power will be utilized during leakage current measurements. The leakage current shall be determined by the voltage-drop method. A true RMS voltmeter shall be used. The voltage shall be measured across the 15 ohm resistor at the highest nominal power line voltage and the highest and lowest nominal power line frequencies for which the equipment is designed. A measurement of 7.5 volts shall correspond to a current of 5 mA. The overall measurement error shall not exceed five percent. The probe shall be used on all external conducting parts such as case, connector housings, recessed calibration or adjustment controls, and control shafts with knobs removed, and the voltage measured for every combination of switch positions available in the test diagram. The open safety ground conductor shall be reconnected immediately after the test is completed.

4.4.3 Environmental inspections. Environmental inspections shall be as listed in TABLE VIII and shall be performed on units which have been subjected to and have passed production inspection.

TABLE VIII. Environmental inspection.

Inspection or test	Requirement paragraph	Test paragraph
Humidity	3.2.4.3	MIL-E-16400
Service shock	3.2.4.2	4.4.3.2
Electromagnetic interference	3.3.3	4.4.3.3

4.4.3.1 Sampling for environmental inspection. One equipment shall be selected from each successive 100 equipments, or fraction thereof, produced. The first sample equipment shall be selected from the first month's production.

4.4.3.2 Service shock environment test. The service shock environment shall be demonstrated by subjecting the equipment to a bench handling test in accordance with MIL-STD-810, Method 516, Procedure V.

4.4.3.3 Electromagnetic interference test procedure. The converter-patch panel shall be tested for electromagnetic interference characteristics in accordance with MIL-STD-462, as required in MIL-STD-461 for Class IC equipment, and shall conform to the requirements of 3.3.3. For production inspection, only tests REO2 and RS01 shall be performed. MIL-HDBK-241 may be used as guidance for EMI reduction in power supplies.

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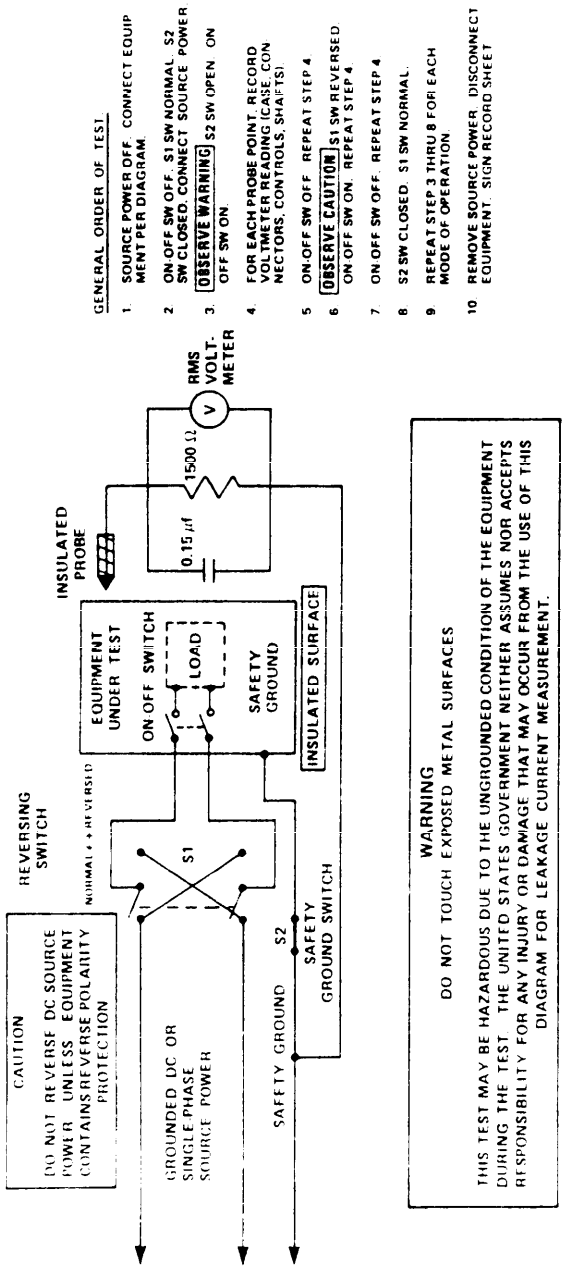


FIGURE 5. Single-phase or DC test diagram for leakage current measurement.

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4.4.3.4 Nonconforming environmental sample units. If a sample unit fails the inspection specified in 4.4.3, the contractor shall immediately investigate the cause of failure and shall report to the quality assurance representative the results thereof and details of the corrective action taken to correct units of product which were manufactured under the same conditions, with the same materials, processes, and so forth. If the quality assurance representative does not consider that the corrective action will enable the product to conform to specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer (see 6.4).

4.4.3.5 Reinspection of conforming environmental sample units. Unless otherwise specified in the contract (see 6.2), sample units which have been subjected to, and have passed, environmental tests may be accepted on the contract provided they are resubjected to, and pass, production inspection after repair of all damage.

4.5 Reliability qualification tests. Reliability qualification tests shall be performed on a minimum of two and a maximum of three converter-patch panel production units. The specific units shall be randomly selected by the quality assurance representative from the first month's production equipments. Testing shall proceed to a decision in accordance with the criteria specified in MIL-STD-781 for Test Plan IV-A. Testing shall be performed at Test Level E of MIL-STD-781.

4.6 Maintainability demonstration. A maintainability demonstration shall be performed to determine the equipment repair time (ERT) in accordance with Method 4 of MIL-STD-471.

4.7 Inspection of preparation for delivery. Inspections shall be conducted to ensure conformance with the requirements of Section 5.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified herein, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing, and marking specified in MIL-E-17555 (see 6.2).

6. NOTES

6.1 Intended use. The converter-patch panel CV-3022/UG is intended for use in digital DC communications systems with the signal data processor AN/SYA-12A, or the data processing set AN/UYK-20(V).

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification
b. Number of first article samples to be submitted if other than specified in 4.3
c. Applicable levels of preservation, packaging, packing, and marking (see 5.1)
d. When reinspected environmental sample units may not be accepted (see 4.4.3.5)

6.3 First article. When a first article is required it shall be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulation (DAR). The first article should be a first production item. The first article should consist of one unit. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, tests, and approval of the first article.

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6.4 Environmental inspection. Approval to ship may be withheld at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action (see 4.4.3.4).

6.4.1 Initial shipments. Initial shipment of end item will be contingent upon successful completion of the environmental inspection.

6.5 Changes from previous issues. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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